SYSEN5680 Term Project: Optimal Parking via Dynamic Programming

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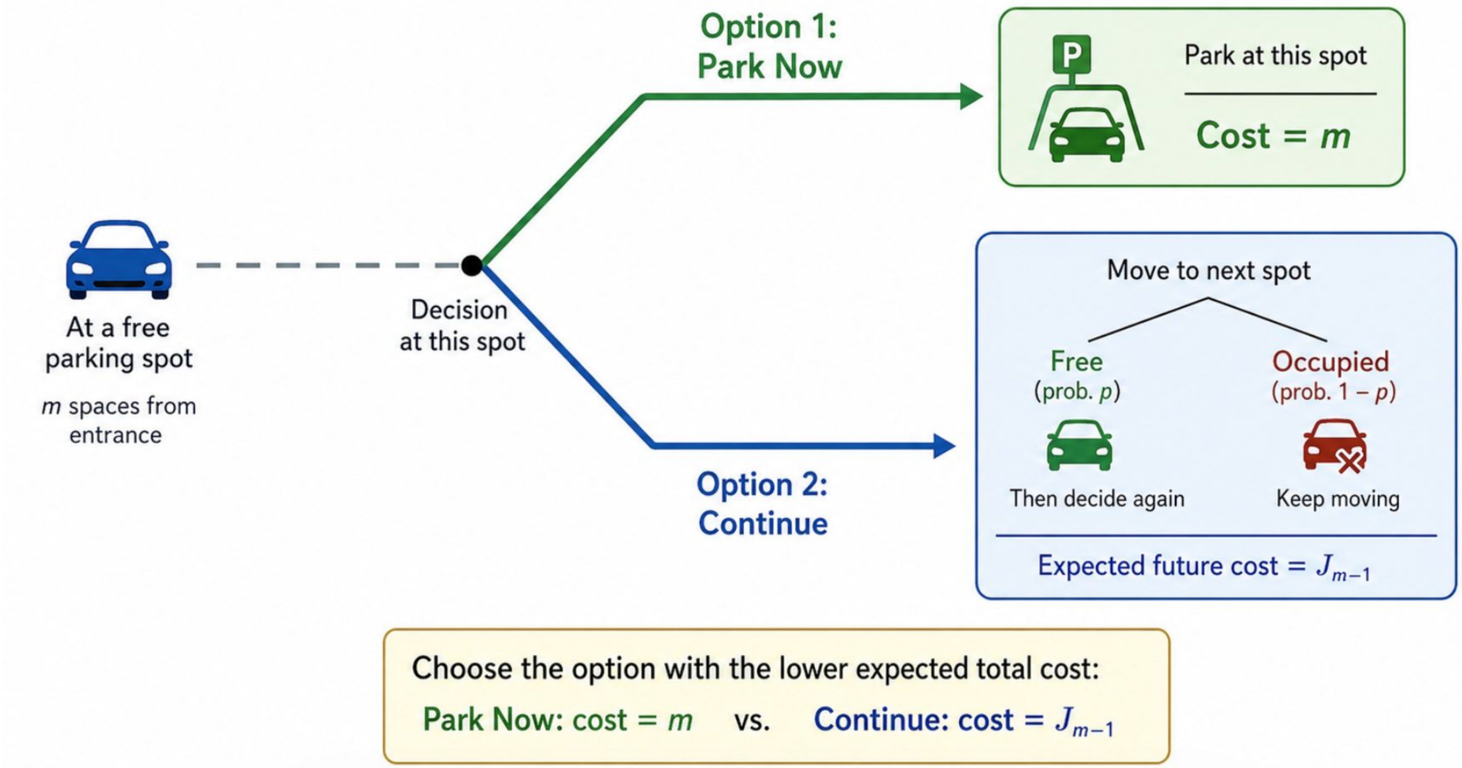
Problem Setup

- Cost = distance m
- If no parking → cost M
- Each spot free with probability p

Goal: minimize distance to entrance under uncertainty

Key Decision Idea



Problem Formulation (MDP Structure)

State

$$X_t = (m_t, W_t)$$

Control

$$U_t \in \{0, 1\}$$

where

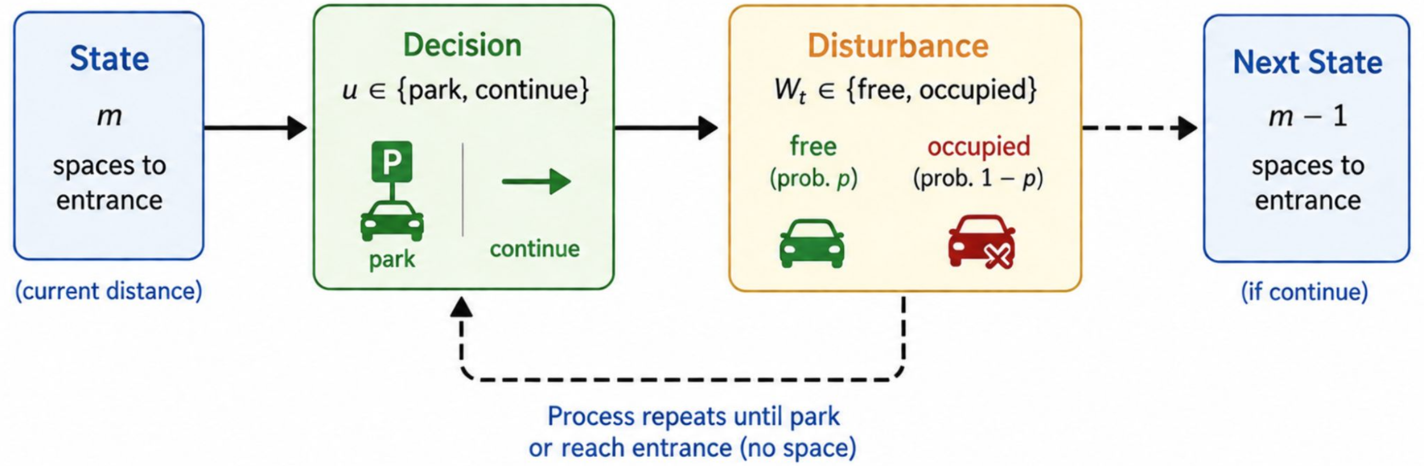
$$U_t = \begin{cases} 1, & \text{park now,} \\ 0, & \text{continue.} \end{cases}$$

Disturbance

$W_t \in \{\text{free}, \text{occupied}\}$
$P(\text{free}) = p$

Cost

Park → m
No parking → M



Dynamic Programming Formulation

Cost-to-go Definition

J_m = optimal expected cost with m spaces remaining

Terminal Condition

$J_0 = M$

(reach entrance without parking)

DP Recursion

$$J_m = \underbrace{(1-p)\,J_{m-1}}_{\substack{\text{occupied (prob. } 1-p)\\ \rightarrow \text{ must continue}}} + \underbrace{p\,\min(m, J_{m-1})}_{\substack{\text{free (prob. } p)\\ \rightarrow \text{ choose best action:}\\ \text{park now or continue}}}$$

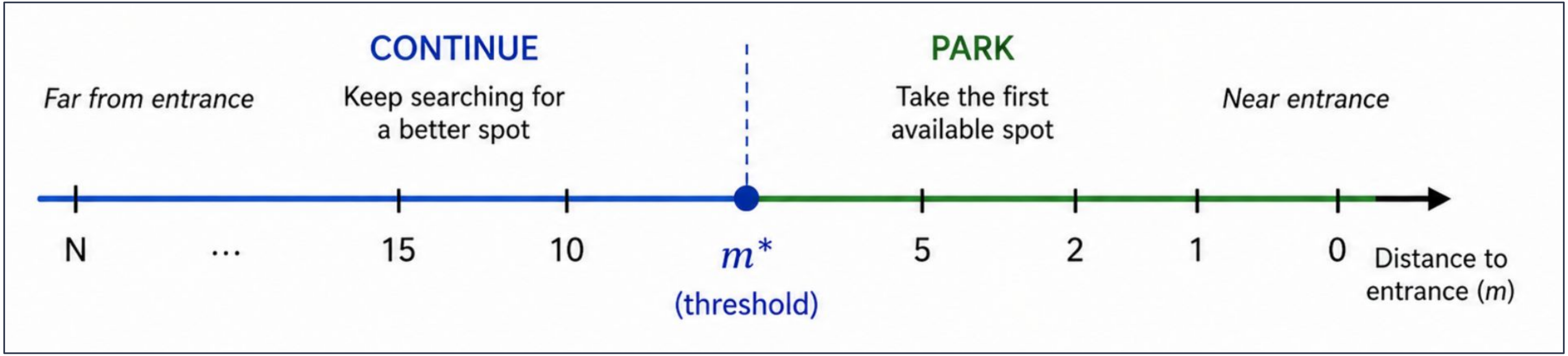
Optimal Decision Rule

$$\mu^*(m) = \begin{cases} \text{park,} & \text{if } m \le J_{m-1} \\ \text{continue,} & \text{otherwise} \end{cases}$$

Park when immediate cost $\le$ expected future cost.

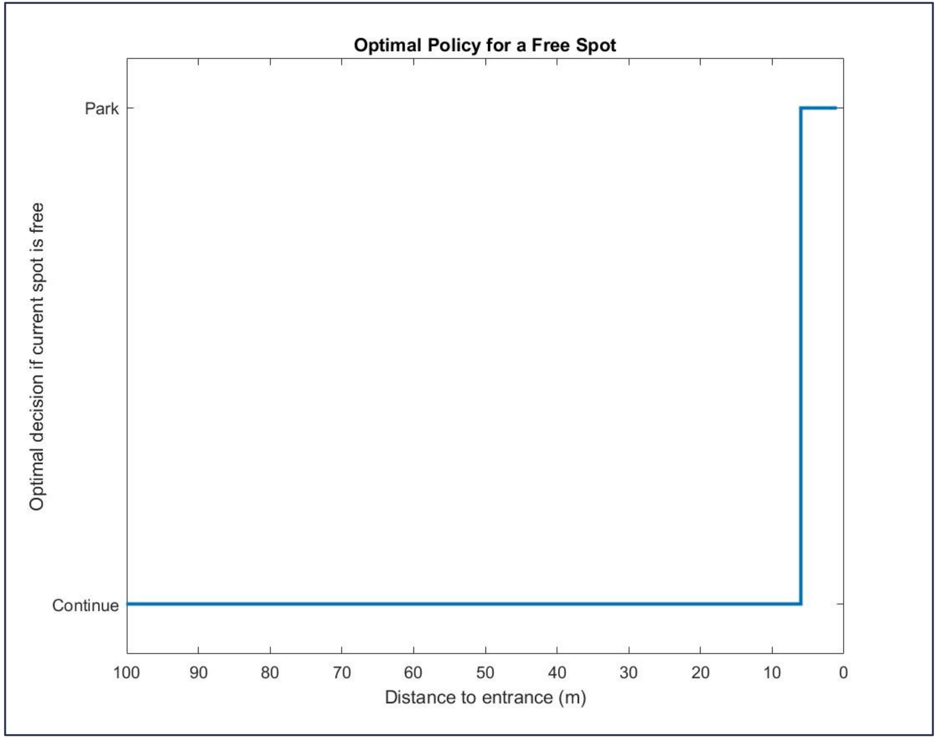
Intuition

There is a critical distance m* that serves as switching point: threshold policy

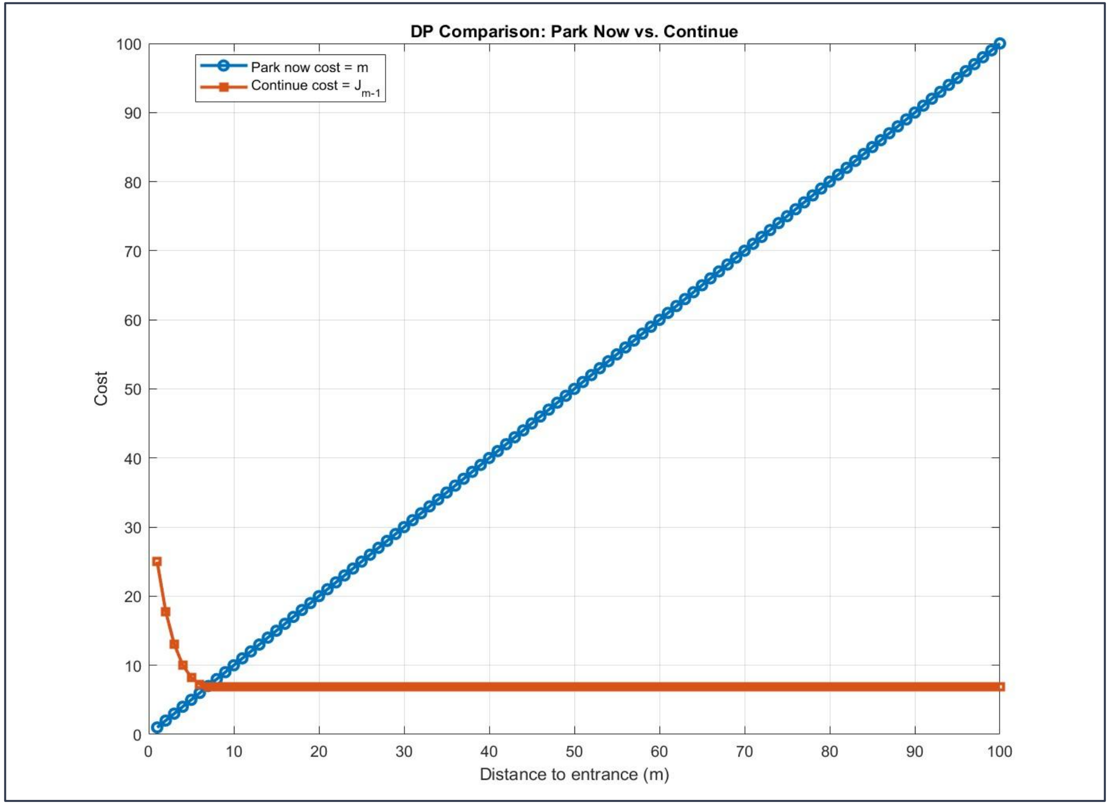


DP Policy Plot

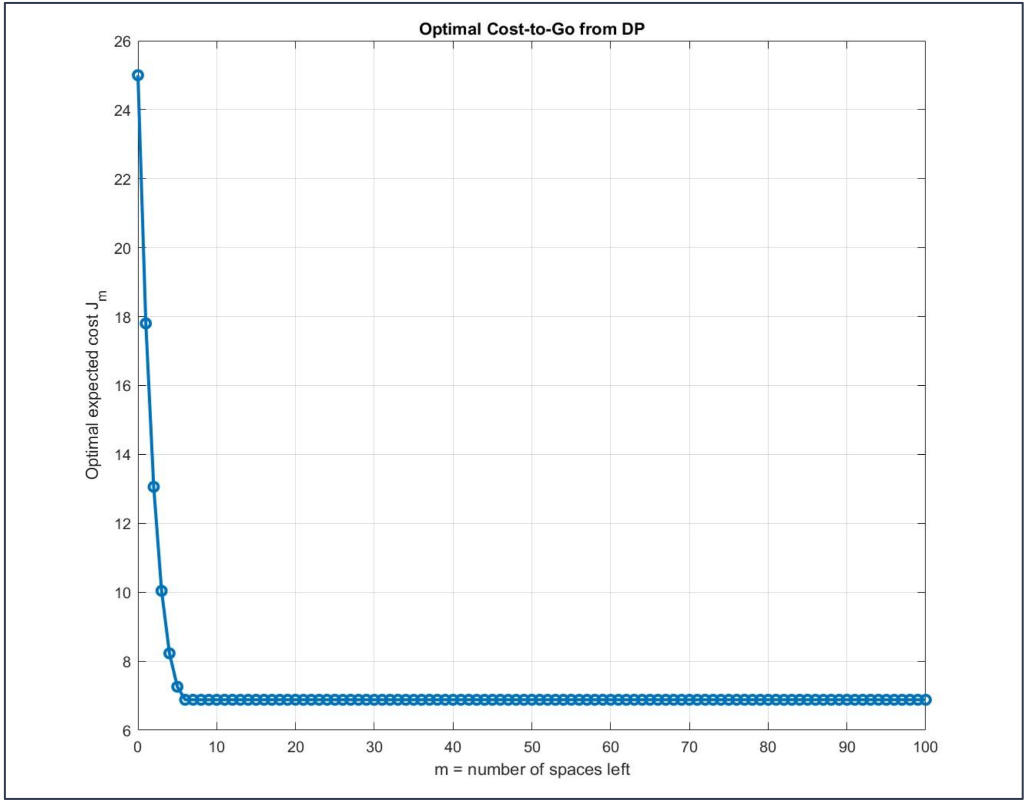
Optimal policy: park when $m \leq m^*$



Cost Comparison (DP Justification)

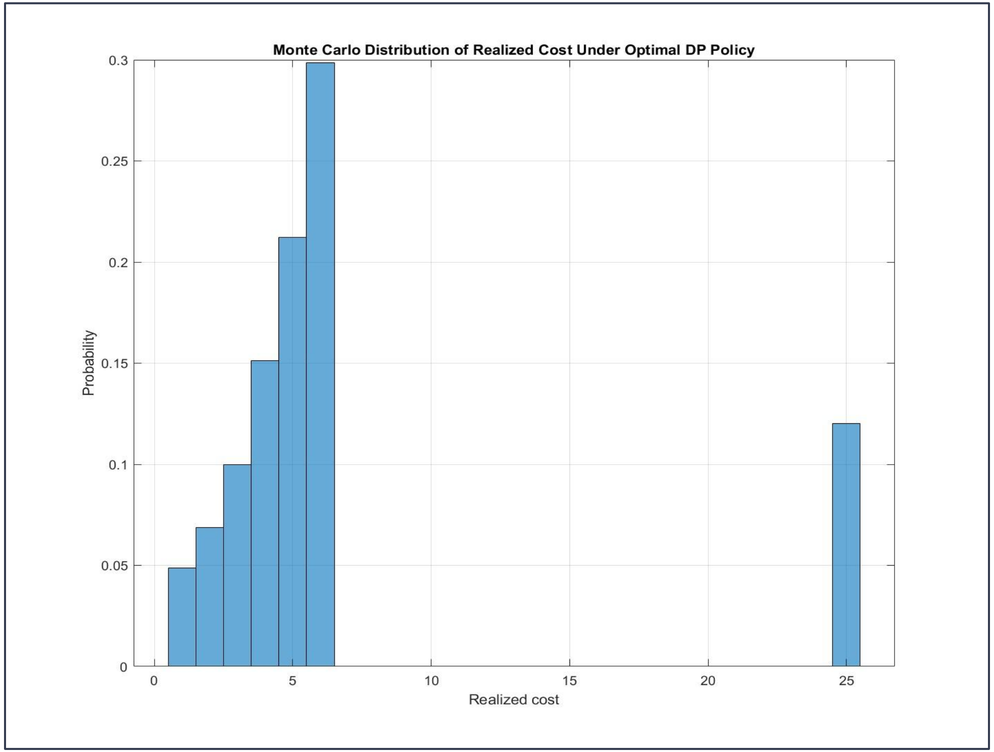


Cost-to-Go Function



Key: Expected cost stabilizes for large m

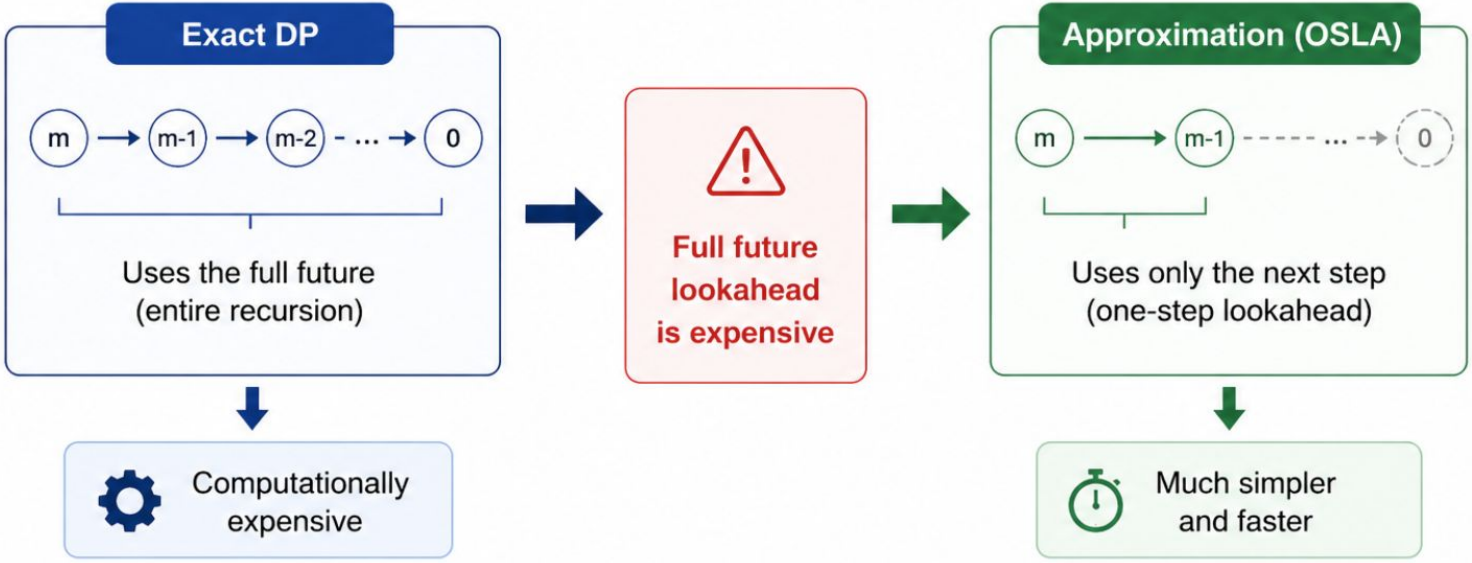
Monte Carlo Validation



Simulation confirms expected cost ≈ DP value

Motivation for Approximation

DP requires full future recursion and can be computationally expensive → approximate by only observing next step

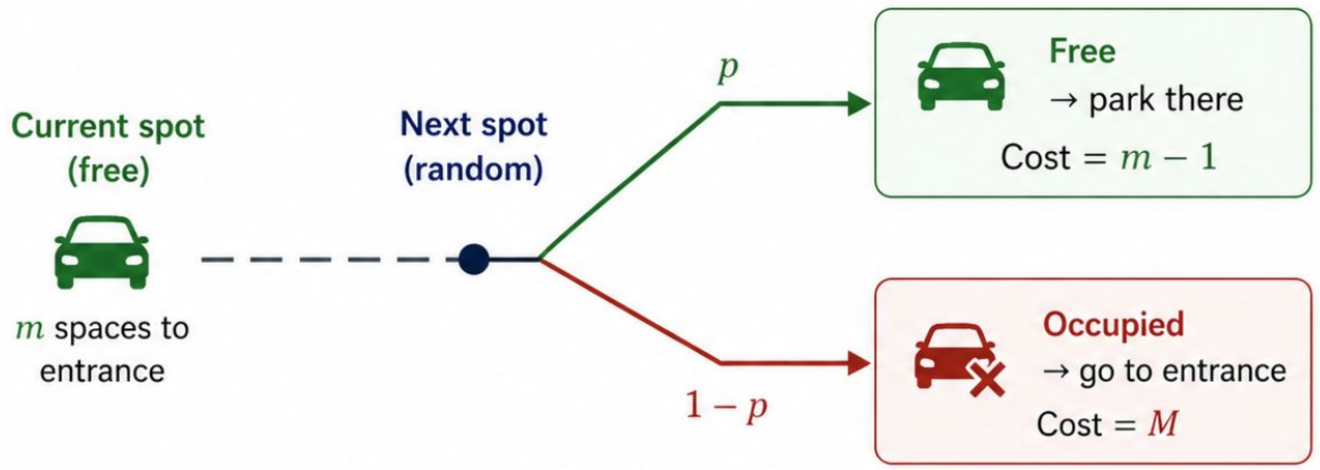


Tradeoff: accuracy vs. simplicity

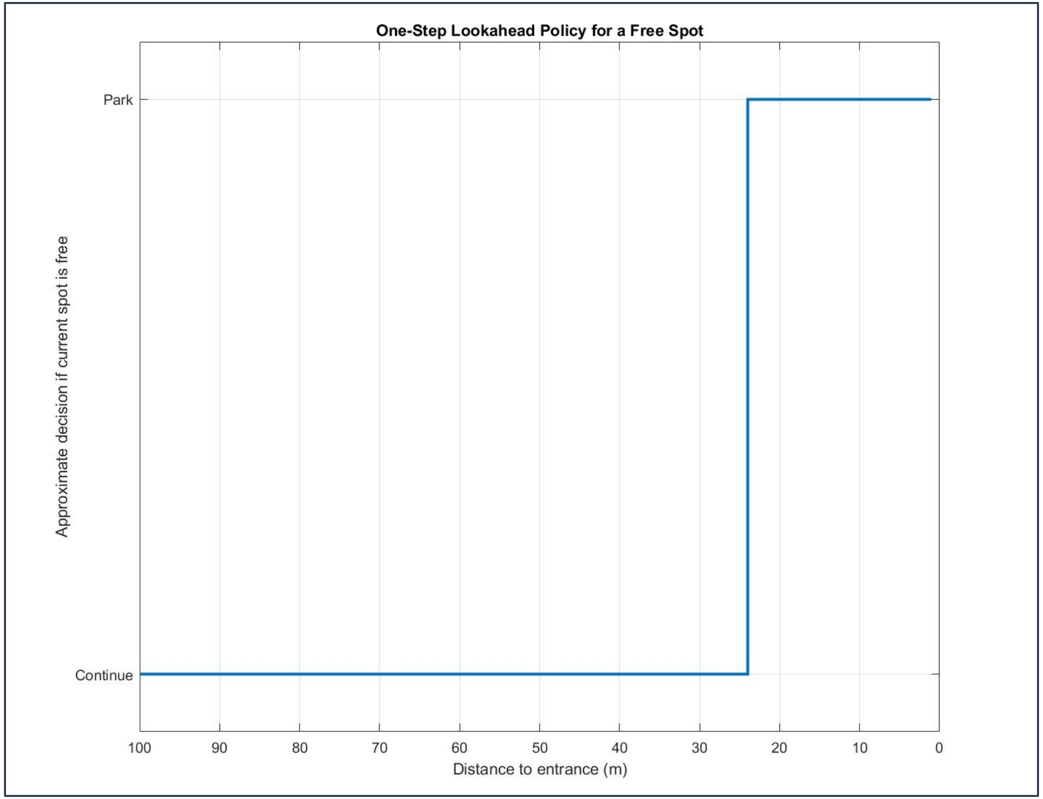
One-Step Lookahead

One-Step Lookahead Continuation Cost:

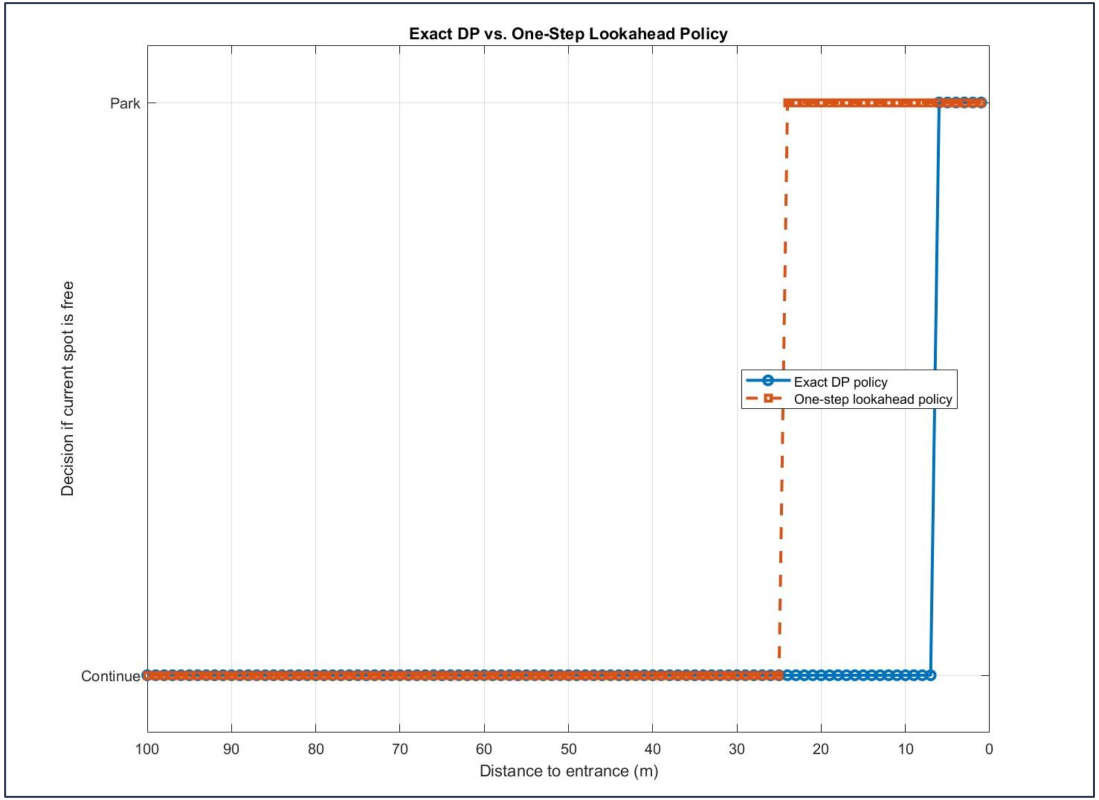
$$\tilde{J}_m = p(m-1) + (1-p)M$$



Approximate Policy Plot

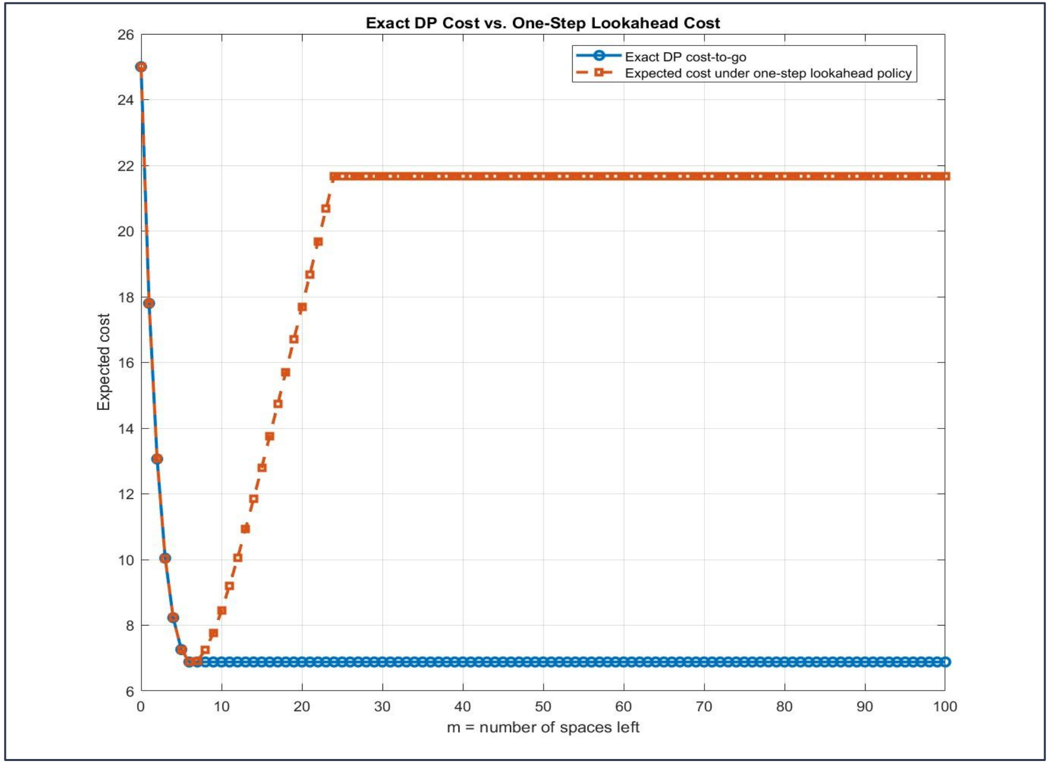


Policy Comparison



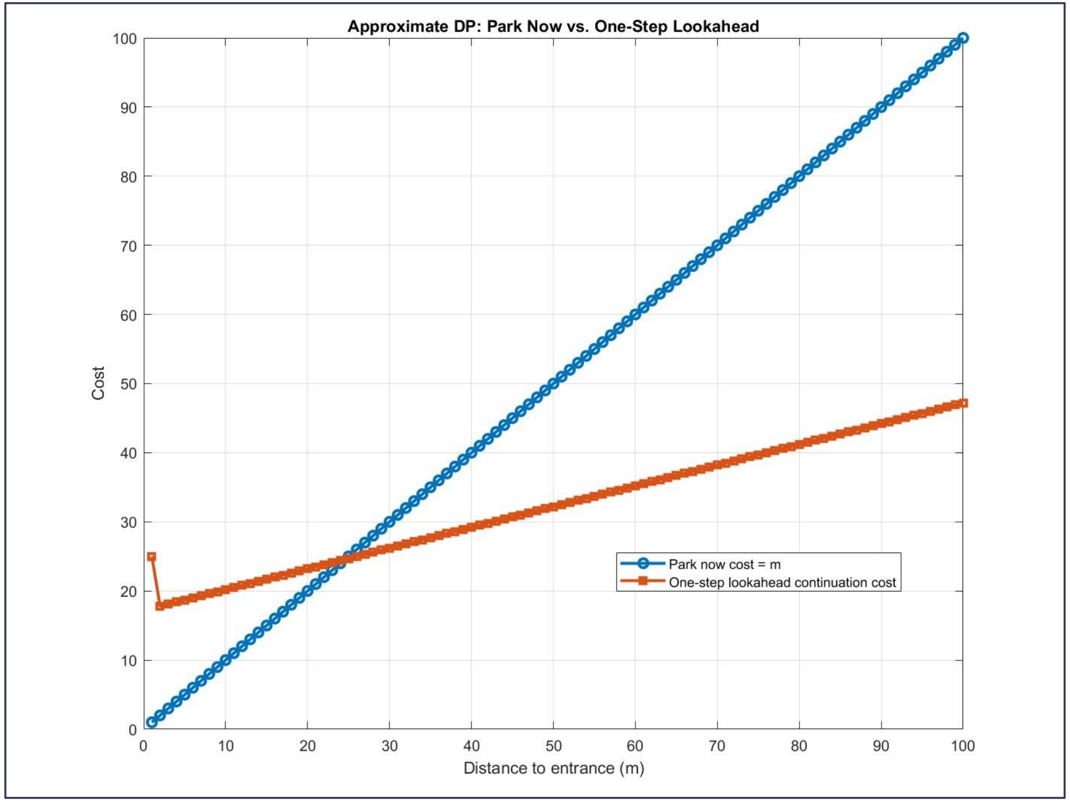
- Exact threshold ≈ 8
- OSLA threshold ≈ 25

Cost Comparison



Large gap → suboptimal

Stop vs Continue (Approximate)



Performance Bound

Exact DP Cost: ~7
Approximate Policy Cost: ~21.5

→

Gap: ~14.5
Relative Error: ~200%

The approximate (one-step lookahead) policy incurs a higher expected cost than the optimal DP policy.

Takeaway: Incorporating future information (DP) leads to substantially better decisions.

Key Takeaways

1. DP → Optimal Policy

DP finds the optimal threshold (m^*).

2. Approximation → Simpler, Not Optimal

One-step lookahead is simple but parks too early
and increases expected cost.

3. Future Information Matters

Full future costs (DP) lead to better decisions
and lower expected cost.

Lookahead and future uncertainty are key to optimal parking.

Thank you!

Questions?